

## Program : Virtual Address

```
GNU nano 8.7.1 VirtualAddress.c
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}
```

## Output :

```
saisharan134@APL-5CD1366K5L:~$ nano VirtualAddress.c
saisharan134@APL-5CD1366K5L:~$ gcc VirtualAddress.c -o VirtualAddress
saisharan134@APL-5CD1366K5L:~$ ./VirtualAddress
Value of x = 10
Virtual address of x = 0x7ffd7b8527f4
saisharan134@APL-5CD1366K5L:~$
```

## Program : Address Space

```
GNU nano 8.7.1 address_space.c
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

    free(heap);

    return 0;
}
```

## Output :

```
saisharan134@APL-5CD1366K5L:~$ nano address_space.c
saisharan134@APL-5CD1366K5L:~$ gcc address_space.c -o address_space
saisharan134@APL-5CD1366K5L:~$ ./address_space
Code address : 0x64e3e20c41a9
Data address : 0x64e3e20c7010
BSS address : 0x64e3e20c7018
Heap address : 0x64e42033e010
Stack address : 0x7fff419cf69c
```

## Program : Demand Paging

```
GNU nano 8.7.1                                demand_paging.c
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

Output :

```
saisharan134@APL-5CD1366K5L:~$ nano demand_paging.c
saisharan134@APL-5CD1366K5L:~$ gcc demand_paging.c -o demand_paging
saisharan134@APL-5CD1366K5L:~$ ./demand_paging
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
saisharan134@APL-5CD1366K5L:~$ |
```

## Program : Address Translation

```
GNU nano 8.7.1                                demand_paging.c
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}
```

Output :

```
saisharan134@APL-5CD1366K5L:~$ nano demand_paging.c
saisharan134@APL-5CD1366K5L:~$ gcc demand_paging.c -o demand_paging
saisharan134@APL-5CD1366K5L:~$ ./demand_paging
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
saisharan134@APL-5CD1366K5L:~$ |
```

Program : Memory Allocation

```
GNU nano 8.7.1 MemoryAllocation.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

    printf("\n");

    free(p);
}
```

Output :

```
saisharan134@APL-5CD1366K5L:~$ nano MemoryAllocation.c
saisharan134@APL-5CD1366K5L:~$ gcc MemoryAllocation.c -o MemoryAllocation
saisharan134@APL-5CD1366K5L:~$ ./MemoryAllocation
10 20 30 40 50
```

## Program : Segmentation Fault

```
GNU nano 8.7.1 SegmentationFault.c
#include <stdio.h>

int main() {

    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}
```

Output :

```
saisharan134@APL-5CD1366K5L:~$ nano SegmentationFault.c
saisharan134@APL-5CD1366K5L:~$ gcc SegmentationFault.c -o SegmentationFault
saisharan134@APL-5CD1366K5L:~$ ./SegmentationFault
Before accessing memory
Segmentation fault ./SegmentationFault
saisharan134@APL-5CD1366K5L:~$ |
```